

# Tait Weicht

Applied Mathematics Ph.D. Candidate

## Contact

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## Education

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|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Ph.D. 2026* | Applied Mathematics<br>University of California, Davis<br>Advisors: Dr. Alexander Wein, Dr. Albert Fannjiang<br><i>*expected graduation date</i> |
| B.S. 2022   | Mathematics                                                                                                                                      |
| B.S. 2022   | Computer Science<br><i>Seattle Pacific University, Seattle WA</i>                                                                                |

## Employment

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020 – Present        | Teaching Assistant<br><i>University of California, Davis</i>                                                                                    |
| Spring 2022 – Present | Graduate Student Researcher<br><i>University of California, Davis</i>                                                                           |
| Summer 2025           | Assistant Instructor<br><i>University of California, Davis</i>                                                                                  |
| Summer 2024           | Teaching Assistant<br>COSMOS Applied Math and Probability Cluster,<br><i>University of California, Davis</i>                                    |
| Summer 2023           | Instructional Materials Contributor (MAT 19 Series A, B and C: Calculus for Data-Driven Applications)<br><i>University of California, Davis</i> |
| Summer 2022           | Assistant Instructor<br><i>University of California, Davis</i>                                                                                  |
| Summer 2021           | C++ Frontend Developer<br><i>WAVES Program at Michigan State University</i>                                                                     |
| Summer 2019           | Frontend Development Intern<br><i>Faithlife Corporation, Bellingham WA</i>                                                                      |

Summer 2018

Student Researcher

*SURIEM at Michigan State University, East Lansing MI*

## Publications and Preprints

### **The Effect of Dose Fractionation on Phase Retrieval in X-Ray Crystallography**

Tait Weicht, Albert Fannjiang

Manuscript in preparation

### **Beyond Frequency Marching: Orbit Recovery in Dihedral and Projected Multireference Alignment**

Tait Weicht, Alexander S. Wein

arXiv preprint 2026, [\[arXiv\]](#)

### **Improving the Threshold for Finding Rank-1 Matrices in a Subspace**

Jeshu Dastidar, Tait Weicht, Alexander S. Wein

arXiv preprint 2025, [\[arXiv\]](#)

## Conferences and Presentations

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | More applications for linear intersections with varieties: projected multi-reference alignment<br><i>Queer in Computational and Applied Mathematics II, Poster Session</i>                   |
| 2026 | Beyond Frequency Marching: Orbit Recovery in Dihedral and Projected Multireference Alignment<br><i>Data and Algebra Seminar, UT Austin</i>                                                   |
| 2024 | More applications for linear intersections with varieties: projected multi-reference alignment<br><i>Foundations of Data Science and Machine Learning at UC4IDS Workshop, Poster session</i> |
| 2024 | Algebraic geometry for cryo-EM: strategies for solving projected multi-reference alignment<br><i>Spectra Survey of Mathematics Conference 2024</i>                                           |
| 2024 | Using projected multi-reference alignment to study challenges in cryo-EM<br><i>Queer in Computational and Applied Mathematics, ICERM</i>                                                     |
| 2023 | Toward orbit recovery for cryogenic electron microscopy: tensor decomposition for projected multi-reference alignment<br><i>ADVANCE Award Symposium, Poster Session</i>                      |

- 2019 Random Walks and the Heat Equation  
*Joint Math Meeting, Student Poster Session*
- 2018 Presentation – Random Walks and the Heat Equation  
*MathFest, Student Paper Session*

## Honors and Awards

- 2020 Graduated summa cum laude from Seattle Pacific University
- 2019 Interdisciplinary Contest in Modeling (ICM) Outstanding Paper Award
- 2018 MAA Outstanding Student Paper Award
- 2018 Mathematical Contest in Modeling (MCM) Meritorious Paper Award
- 2016 – 2020 President's List

## Professional Memberships

Society of Industrial and Applied Mathematics ([SIAM](#))  
 American Mathematical Society (AMS)  
 The Association for LGBTQ+ Mathematicians ([Spectra](#))

## Associate Instructorships

- Summer 2025 Linear Algebra ([MAT 22A](#)), UC Davis
- Summer 2021 Calculus for Biology and Medicine ([MAT 17B](#)), UC Davis

## Teaching Assistantships

- Spring 2025 Calculus for Biology and Medicine ([MAT 17B](#)), UC Davis
- Winter 2025 Probability ([MAT 135](#)), UC Davis
- Fall 2024 Probability ([MAT 135](#)), UC Davis
- Winter 2024 Calculus for Data-Driven Applications ([MAT 19B](#)), UC Davis
- Fall 2023 Calculus for Data-Driven Applications ([MAT 19A](#)), UC Davis
- Spring 2023 Linear Algebra ([MAT 22A](#)), UC Davis
- Winter 2023 Introduction to Abstract Math ([MAT 108](#)), UC Davis
- Fall 2022 Calculus: Differential Calculus ([MAT 21A](#)), UC Davis

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| Spring 2022 | Calculus for Biology and Medicine ( <a href="#">MAT 17C</a> ), <i>UC Davis</i>        |
| Winter 2022 | Calculus for Biology and Medicine ( <a href="#">MAT 17B</a> ), <i>UC Davis</i>        |
| Fall 2021   | Calculus for Biology and Medicine ( <a href="#">MAT 17A</a> ), <i>UC Davis</i>        |
| Spring 2021 | Calculus: Partial Derivatives and Series ( <a href="#">MAT 21C</a> ), <i>UC Davis</i> |
| Winter 2021 | Calculus: Partial Derivatives and Series (MAT 21C), <i>UC Davis</i>                   |
| Fall 2020   | Calculus: Partial Derivatives and Series (MAT 21C), <i>UC Davis</i>                   |

### Departmental Service

|                            |                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------|
| Fall 2025 –<br>Spring 2026 | Secretary for UC Davis’s Spectra chapter,<br><i>UC Davis</i>                                 |
| Fall 2024 –<br>Spring 2025 | President for UC Davis’s Spectra chapter,<br><i>UC Davis</i>                                 |
| Spring 2024                | Social Chair for UC Davis <a href="#">Galois Group</a> (grad student group)                  |
| Winter 2024                | Graduate Mentor for the Directed Reading Program ( <a href="#">DRP</a> ),<br><i>UC Davis</i> |
| Winter 2022 –<br>Fall 2023 | Graduate Mentor for the Directed Reading Program (DRP),<br><i>UC Davis</i>                   |

### Software Development

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|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fall 2023 –<br>Ongoing | Efficient recursive method for recovering orbits in dihedral multi-reference alignment (MRA) and MRA with projection (original work)                                                               |
| 2023 –<br>Ongoing      | Compute linear intersections with Segre-Veronese varieties + Jennrich’s algorithm to decompose real or complex-valued tensors<br>( <a href="#">implementation</a> , <a href="#">source paper</a> ) |
| Fall 2023              | <a href="#">ESIIL Hackathon</a> participant, work with OpenStreetMaps in R and geospatial data ( <a href="#">solution repo</a> , <a href="#">my contributions</a> )                                |
| Spring 2023            | Algorithm to solve linear systems with a canonical polyadic decomposition-constrained solution<br>( <a href="#">source paper</a> , DOI: 10.1002/nla.2190)                                          |
| Fall 2022              | Spectral synchronization algorithm for cryo-EM common-line data ( <a href="#">source paper</a> , DOI: 10.1137/090767777)                                                                           |

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| Summer 2021 | C++ code for the <a href="#">Empirical</a> library's prefabricated web components as part of the WAVES summer program |
| Fall 2019   | Spatial hashing and physics updates for flocking simulator ( <a href="#">project here</a> )                           |
| Winter 2018 | Turing-machine-on-a-graph interpreter program (inspired by <a href="#">source paper</a> )                             |
| Fall 2017   | Kohonen's self-organizing map algorithm for dimensionality reduction                                                  |

## References

### Research

Dr. Alexander Wein (UC Davis)

[aswein@ucdavis.edu](mailto:aswein@ucdavis.edu)

Dr. Albert Fannjiang (UC Davis)

[fannjiang@math.ucdavis.edu](mailto:fannjiang@math.ucdavis.edu)

### Teaching

Dr. Rohit Thomas (UC Davis)

[rthomas@math.ucdavis.edu](mailto:rthomas@math.ucdavis.edu)

Dr. Korana Burke (UC Davis)

[kburke@ucdavis.edu](mailto:kburke@ucdavis.edu)